9.3 Designing the future architecture of a data product and related systems

In the previous section, we showed you the process of capturing current architecture and collecting architectural drivers, through the example of the Cost Statement data product. In this section, we will follow up on that example to show you the process of designing the data product architecture. We will not limit ourselves to only one example. The following sections present multiple examples from Messflix, and we will try to touch on as many commonly used architectural patterns as possible. We will discuss the following cases:

- File-based data product (spreadsheet)
- Existing monolithic and microservice architecture as a source of data for a data product
- Stream- and batch-processing examples

In each example, we will analyze architectural drivers, possible solutions, and tradeoffs related to these solutions. While reading, don't focus too much on the specific technologies used (e.g., MongoDB); they are provided just as examples. Now let's delve into what the design activity looks like.

9.3.1 Design session

We strongly believe that design should be a collaborative and collective effort done by the whole or by most of the development team. This helps avoid the pitfall of the ivory tower architect.

DEFINITION An *ivory tower architect* is a software architect who is isolated from others, mainly the development team, because of organizational structure, company culture, or personal approach.

When everyone is involved in the design of a data product, the team will feel connected to it; it will be their child. This is why you should invite the whole team to the design session. During the session, it is important to make the exercise inclusive and allow each team member to have a voice.